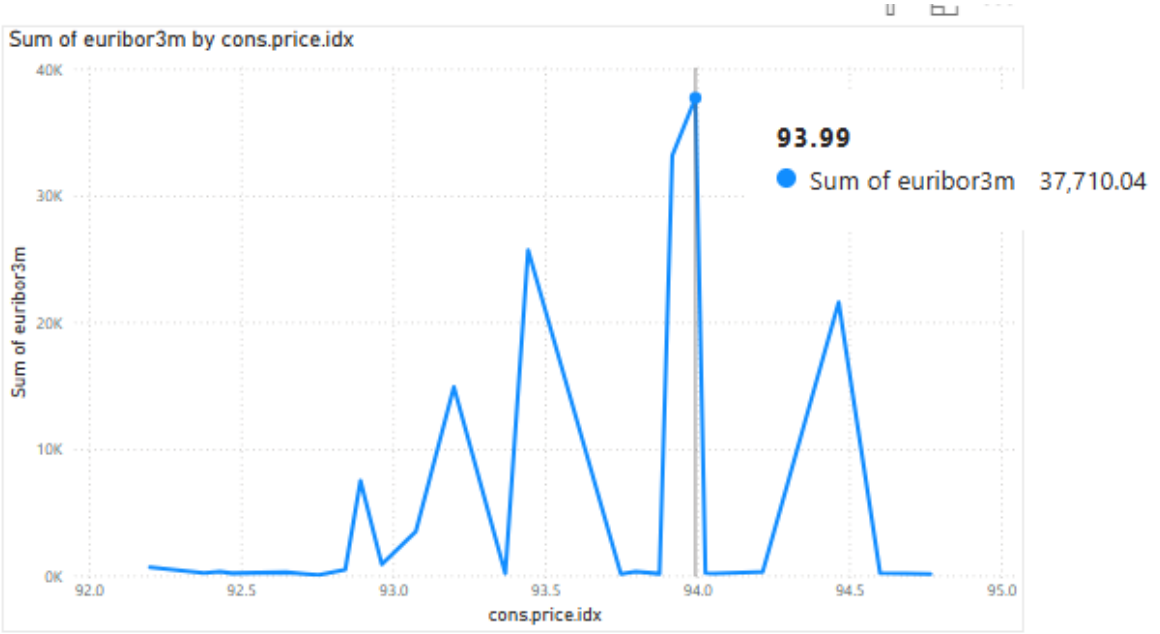
Data Cleansing, Dashboard Design & Visualization

a)

This dataset is based on "Bank Marketing" UCI dataset published by the Banco de Portugal and takes into consideration privacy concerns hence why it is ideal for analysis. The goal is to analyze the Bank Marketing dataset to understand factors influencing term deposit subscriptions ('variable y') and subsequently build a model to predict if a customer will subscribe. The dataset contains 41,188 rows and 21 columns, of which 10 columns are numerical, and 11 columns are categorical.

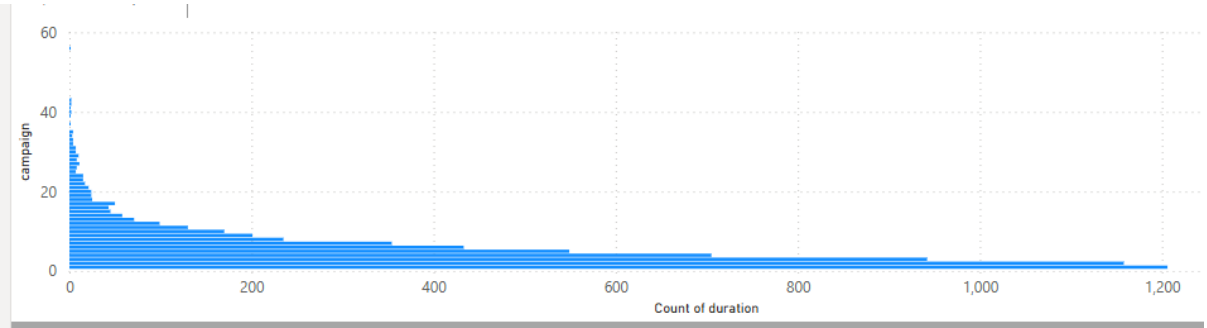
Numeric columns include Age, Duration ,Campaign, PDays, Previous, emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m and nr.employed .For Age the minimum age is 17 years and the maximum age is 98 years with the mean being 40years of age & median being 38 years of age. Campaign is the total number of contacts made with the specific client during the current campaign and the minimum is 1 day while the maximum is 4 days, having the mean as 2.57 and the median as 2.
PDays represents the Number of days elapsed since the client was last contacted during a previous campaign (a value of 999 indicates the client has not been contacted before). Previous represents the number of contacts made with this client prior to this campaign. Quarterly indicators include nr.employed which is number of employees, emp.var.rate stands for employment variation rate. Monthly indicators include cons.price.idx which stands for consumer price index and measures inflating changes in the price of a basket of good and services while cons.conf.idx stands for consumer confidence index which reflects the view of the public regarding the economy with higher values signifying more optimism and lower values signifying pessimism . euribor3m is euribor 3 month rate and is a daily indicator which is the average rate at which European Banks lend money to each other. (UCI Machine Learning Repository, 2025). **GRAPH 1 BELOW**



The above line graph **[Graph 1]** is the sum of euribor3m by cons.price.idx .This graph indicates that The two are directly proportional and the max sum the euribor3m got is 37,710 .A possibility may be that Banks adjust the interest they receive from borrowing other banks money based on the current inflation in order to counter inflation and also ensure targeted profit margin is still reached regardless the economy as banks are businesses.

Duration is the duration of the last contact with the client in seconds and the duration is not known until after the call is made, Once the call concludes, the outcome y becomes clear.

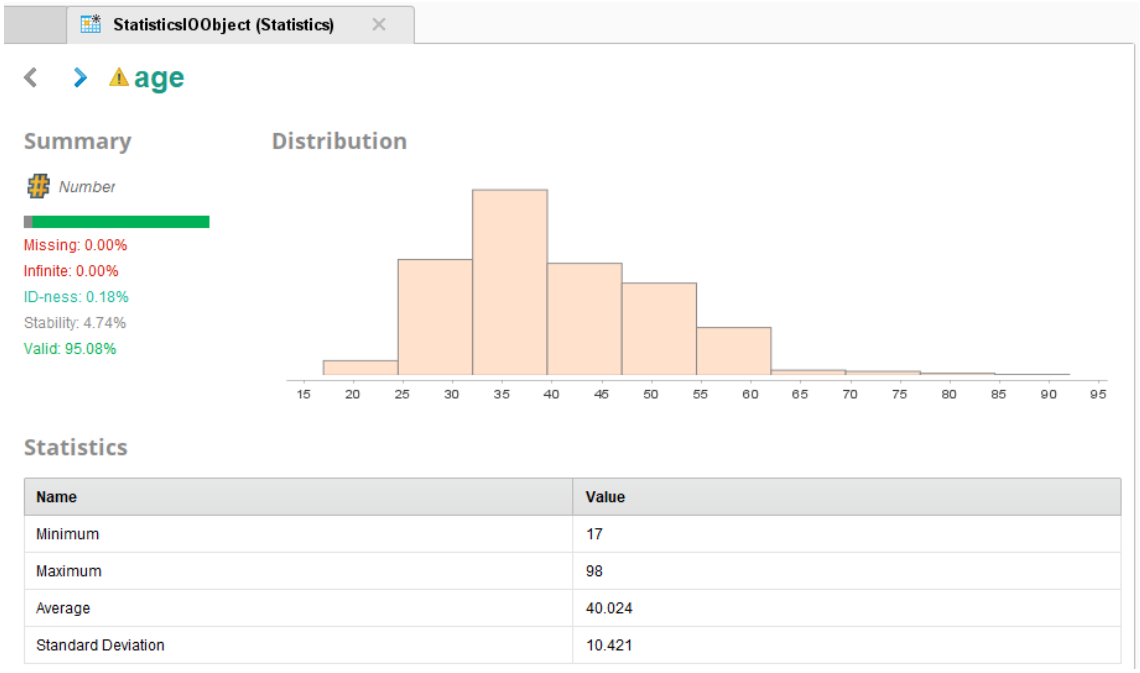
Graph 2



campaign	Count of duration
1	1206
2	1158
3	942
4	705
5	549
6	433
7	354
8	235
9	201
10	170

The above table **[Graph 2]** shows count of duration by campaign. 1 shows that the client was contacted once and there are more seconds spent on the clients on the first call which may be due to eagerness in wanting to know what the campaign is all about. The more times a customer is called the less seconds spent which may simply showcase how customers easily get bored per visit and are eager to get off the call as they may not be hearing anything new that may interest them. The table goes up to 56 which has count of duration as 1.

Statistical representations/summaries of the explanations of the numeric attributes are attached below and were obtained using the rapid miner statistics tab:



Name	Value
Minimum	17
Maximum	98
Average	40.024
Standard Deviation	10.421

StatisticsIOObject (Statistics)

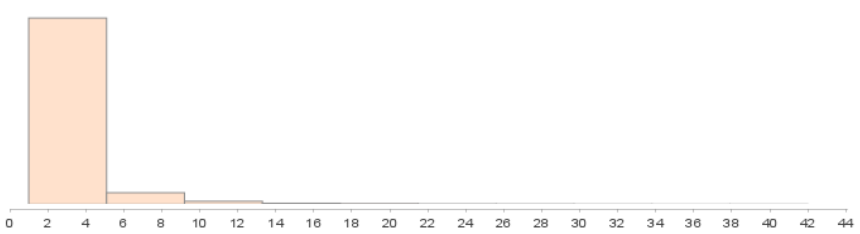
campaign

Summary

Number

Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.09%
Stability: 42.28%
Valid: 57.63%

Distribution



Statistics

Name	Value
Minimum	1
Maximum	56
Average	2.568
Standard Deviation	2.770

StatisticsIOObject (Statistics)

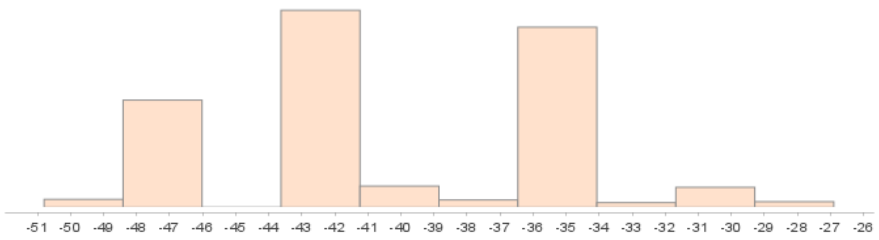
cons.conf.idx

Summary

Number

Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.01%
Stability: 18.45%
Valid: 81.54%

Distribution



Statistics

Name	Value
Minimum	-50.800
Maximum	-26.900
Average	-40.503
Standard Deviation	4.628

StatisticsIOObject (Statistics)

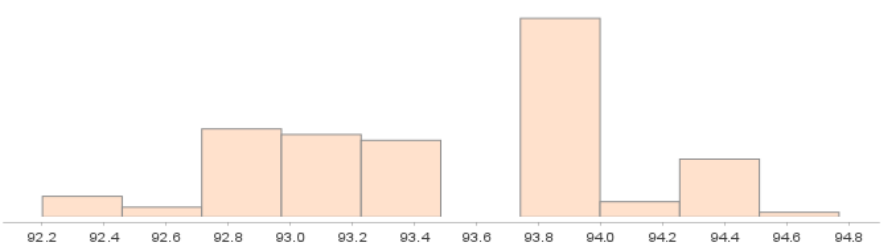
cons.price.idx

Summary

Number

Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.00%
Stability: 18.45%
Valid: 81.55%

Distribution



Statistics

Name	Value
Minimum	92.201
Maximum	94.767
Average	93.576
Standard Deviation	0.579

StatisticsIOObject (Statistics)

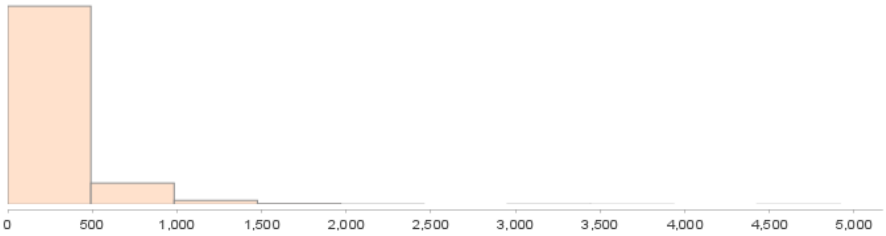
duration

Summary

Number

Missing: 0.00%
Infinite: 0.00%
ID-ness: 2.61%
Stability: 0.53%
Valid: 96.86%

Distribution



Statistics

Name	Value
Minimum	0
Maximum	4918
Average	258.285
Standard Deviation	259.279

StatisticsIOObject (Statistics)

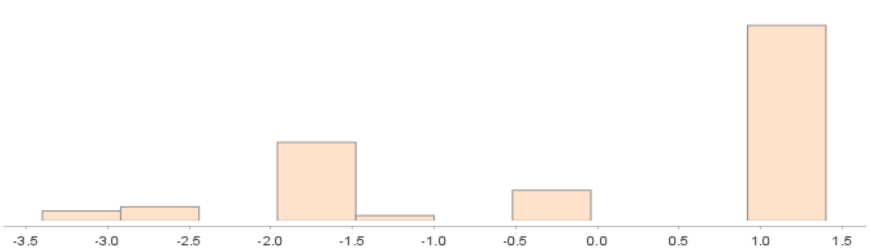
emp.var.rate

Summary

Number

Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.00%
Stability: 39.98%
Valid: 60.01%

Distribution



Statistics

Name	Value
Minimum	-3.400
Maximum	1.400
Average	0.082
Standard Deviation	1.571

StatisticsIOObject (Statistics)

euribor3m

Summary

Number

Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.00%
Stability: 6.82%
Valid: 93.18%

Distribution



Statistics

Name	Value
Minimum	0.634
Maximum	5.045
Average	3.621
Standard Deviation	1.734

StatisticsIOObject (Statistics)

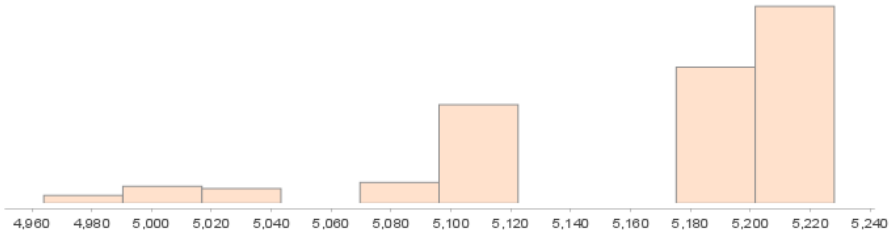
nr.employed

Summary

Number

Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.03%
Stability: 39.98%
Valid: 59.99%

Distribution



Statistics

Name	Value
Minimum	4964
Maximum	5228
Average	5167.019
Standard Deviation	72.178

StatisticsIOObject (Statistics)

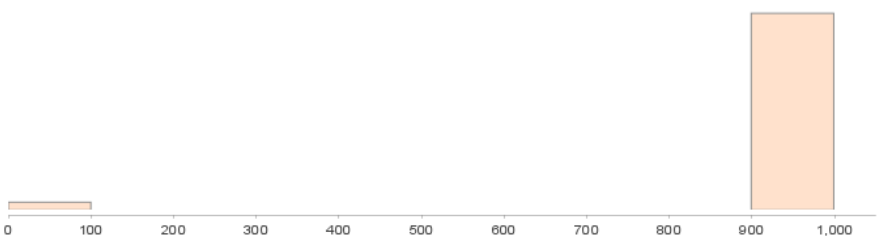
pdays

Summary

Number

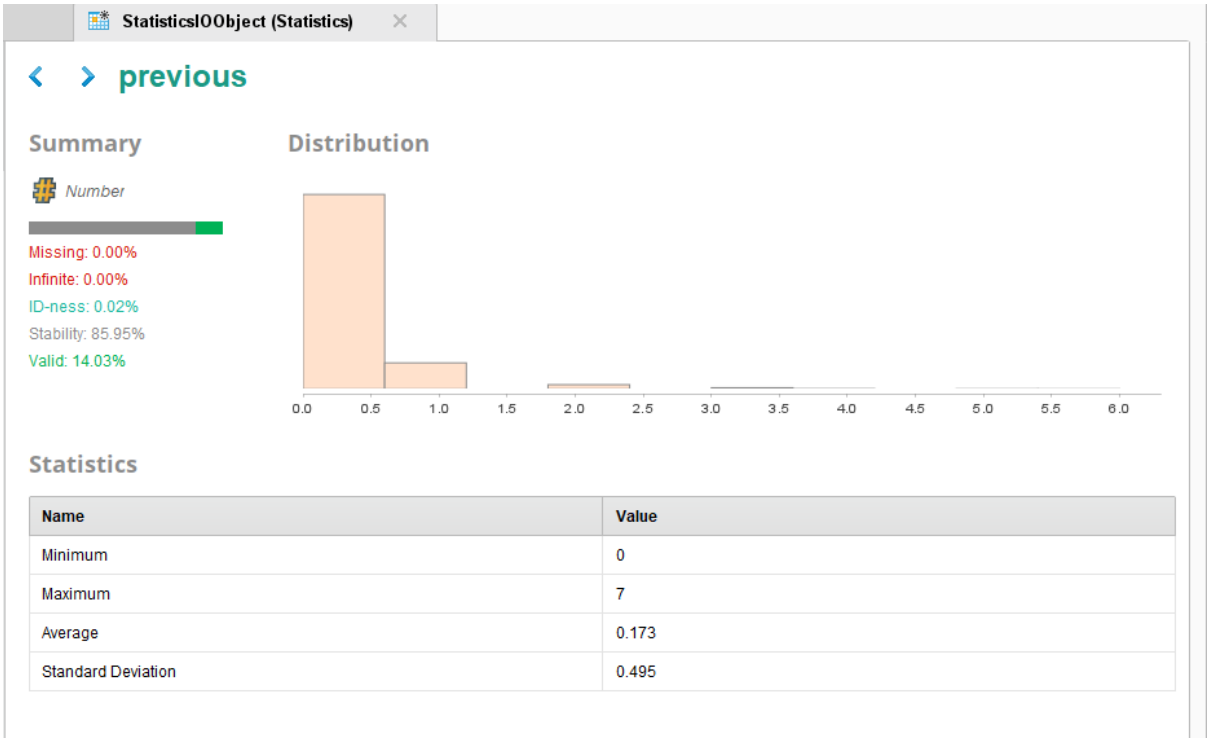
Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.06%
Stability: 96.18%
Valid: 3.76%

Distribution



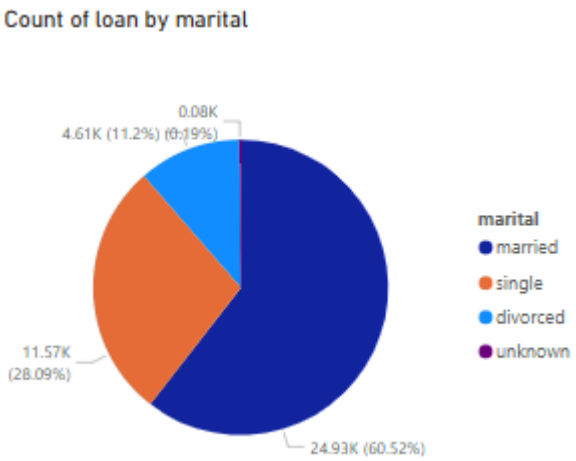
Statistics

Name	Value
Minimum	0
Maximum	999
Average	962.475
Standard Deviation	186.911



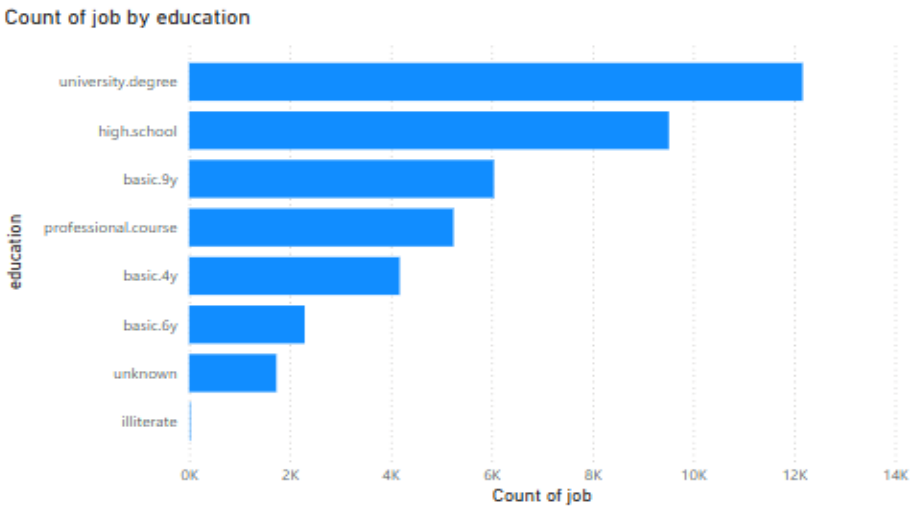
Name	Value
Minimum	0
Maximum	7
Average	0.173
Standard Deviation	0.495

Categorical columns include Job, Marital ,Education , Loan, Default, Housing ,Contact ,Month ,day of week ,poutcome and y . Default is whether the customer has credit by default and only 3 clients had credit by default and 8597 were unknown, leaving 32588 who did not have credit by default. Housing is for if the customer has a housing loan and loan is for whether the customer has a Personal loan. Contact signifies contact communication type example Telephone and Cellular , Month signifies the last month the were contacted in the year while day of week is for the day they were last contacted in the week. Poutcome represents what happened from the previously held campaign and y is the final desired target which is whether the client subscribed a term deposit after the campaign. (UCI Machine Learning Repository, 2025)



Graph 3

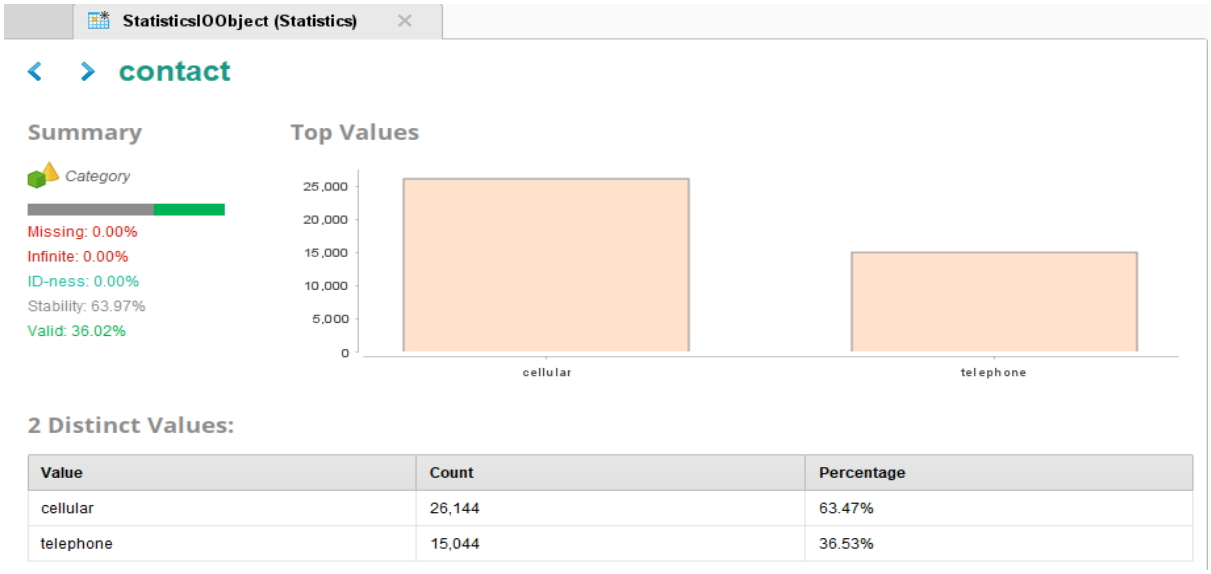
The above pie chart **[Graph 3]** is of Count of loan by Marital status and it shows that Married people take more Personal Loans from banks with their count being at 24.93K and taking up 60.52% of the pie chart. This could be because in a marriage there is pressure to take care of families and live a certain lifestyle that would make the family seem ideal to outsiders.



Graph 4

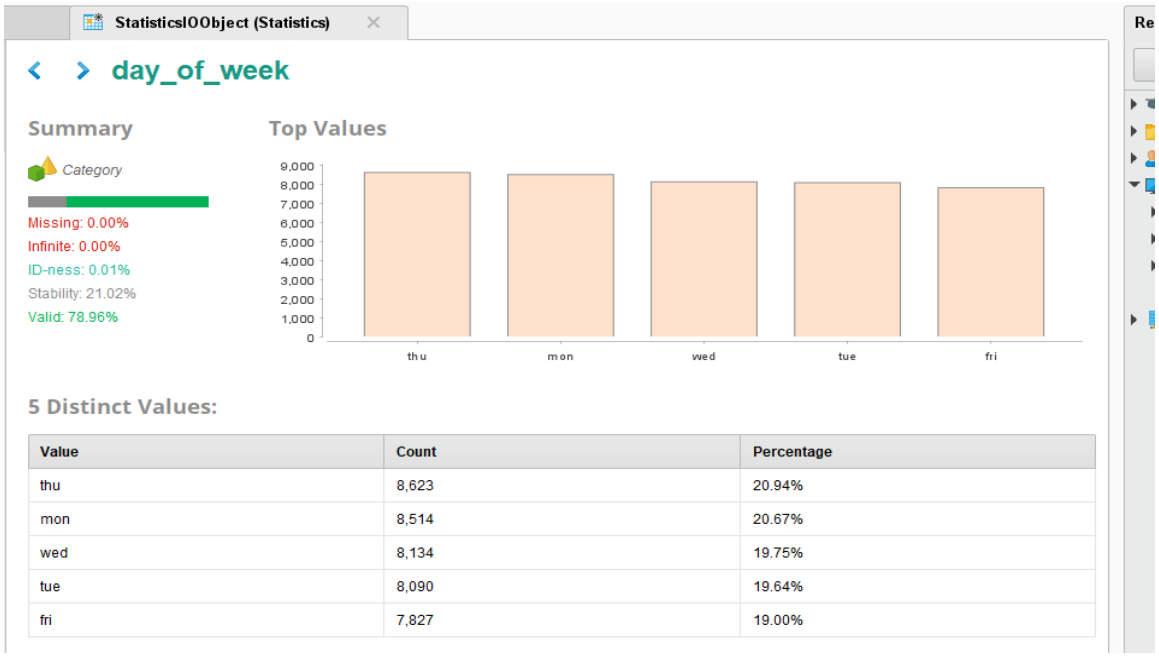
The above Stacked Bar Chart **[Graph 4]** shows that Educated people are more likely to find jobs than the illiterate ,with the count of those with university degrees employed being 12168 and the illiterate employed being only 18. Those employed with Highschool qualifications are 9515,those employed with basic 9 years of education are 6045, those who did professional course and are employed are 5243, those with basic 4 years education employed are 4176 ,those employed with basic 6 years education are 2292 which may be classified as abnormal as you would expect more years of education to equal higher chances of finding a job .The unknowns are to be discarded as we cannot guess someone's education.

Statistical representations/summaries of the explanations of the categorical attributes are attached below and were obtained using the rapid miner statistics tab:



2 Distinct Values:

Value	Count	Percentage
cellular	26,144	63.47%
telephone	15,044	36.53%



5 Distinct Values:

Value	Count	Percentage
thu	8,623	20.94%
mon	8,514	20.67%
wed	8,134	19.75%
tue	8,090	19.64%
fri	7,827	19.00%

StatisticsIOObject (Statistics)

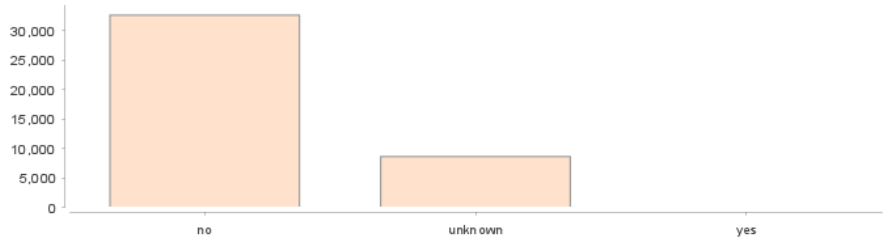
default

Summary

Category

Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.00%
Stability: 79.25%
Valid: 20.74%

Top Values



3 Distinct Values:

Value	Count	Percentage
no	32,588	79.12%
unknown	8,597	20.87%
yes	3	0.01%

StatisticsIOObject (Statistics)

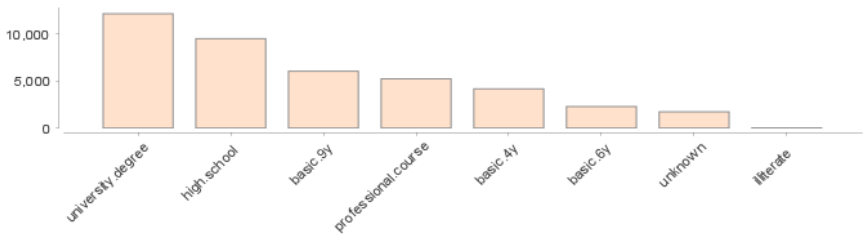
education

Summary

Category

Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.02%
Stability: 30.34%
Valid: 69.64%

Top Values



8 Distinct Values:

Value	Count	Percentage
university.degree	12,168	29.54%
high.school	9,515	23.10%
basic.9y	6,045	14.68%
professional.course	5,243	12.73%
basic.4y	4,176	10.14%
basic.6y	2,292	5.56%

StatisticsIOObject (Statistics)

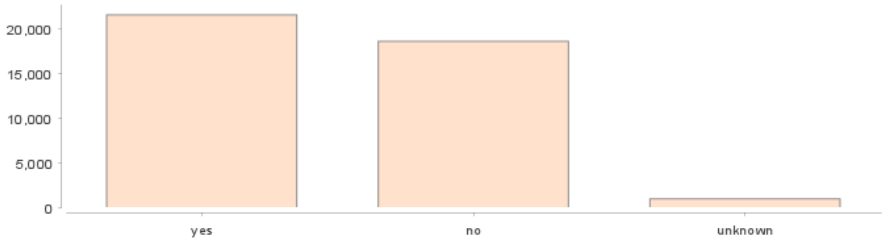
housing

Summary

Category

Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.01%
Stability: 52.48%
Valid: 47.52%

Top Values



3 Distinct Values:

Value	Count	Percentage
yes	21,576	52.38%
no	18,622	45.21%
unknown	990	2.40%

StatisticsIOObject (Statistics)

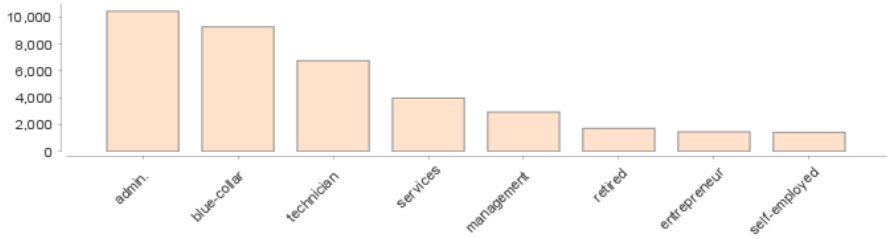
job

Summary

Category

Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.03%
Stability: 26.31%
Valid: 73.66%

Top Values



12 Distinct Values:

Value	Count	Percentage
admin.	10,422	25.30%
blue-collar	9,254	22.47%
technician	6,743	16.37%
services	3,969	9.64%
management	2,924	7.10%

StatisticsIOObject (Statistics)

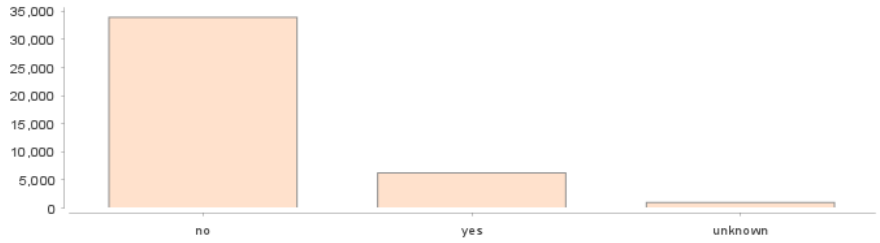
loan

Summary

Category

Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.01%
Stability: 82.85%
Valid: 17.15%

Top Values



3 Distinct Values:

Value	Count	Percentage
no	33,950	82.43%
yes	6,248	15.17%
unknown	990	2.40%

StatisticsIOObject (Statistics)

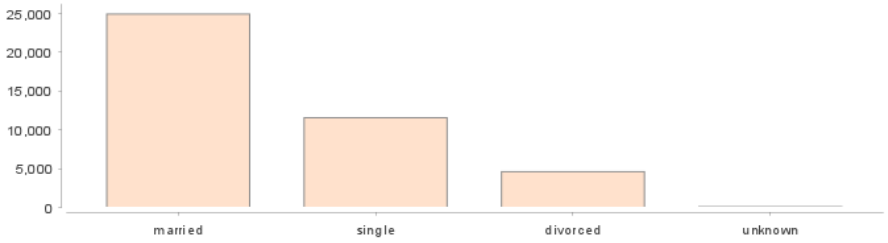
marital

Summary

Category

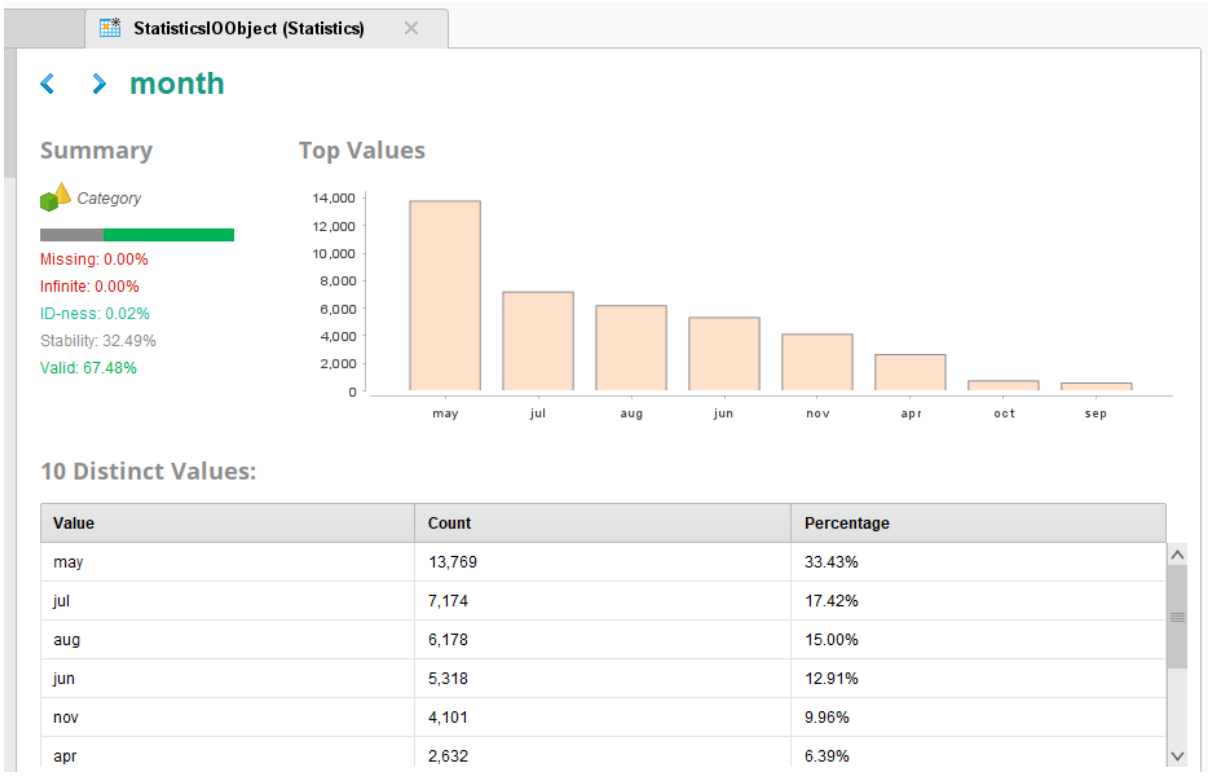
Missing: 0.00%
Infinite: 0.00%
ID-ness: 0.01%
Stability: 60.25%
Valid: 39.74%

Top Values



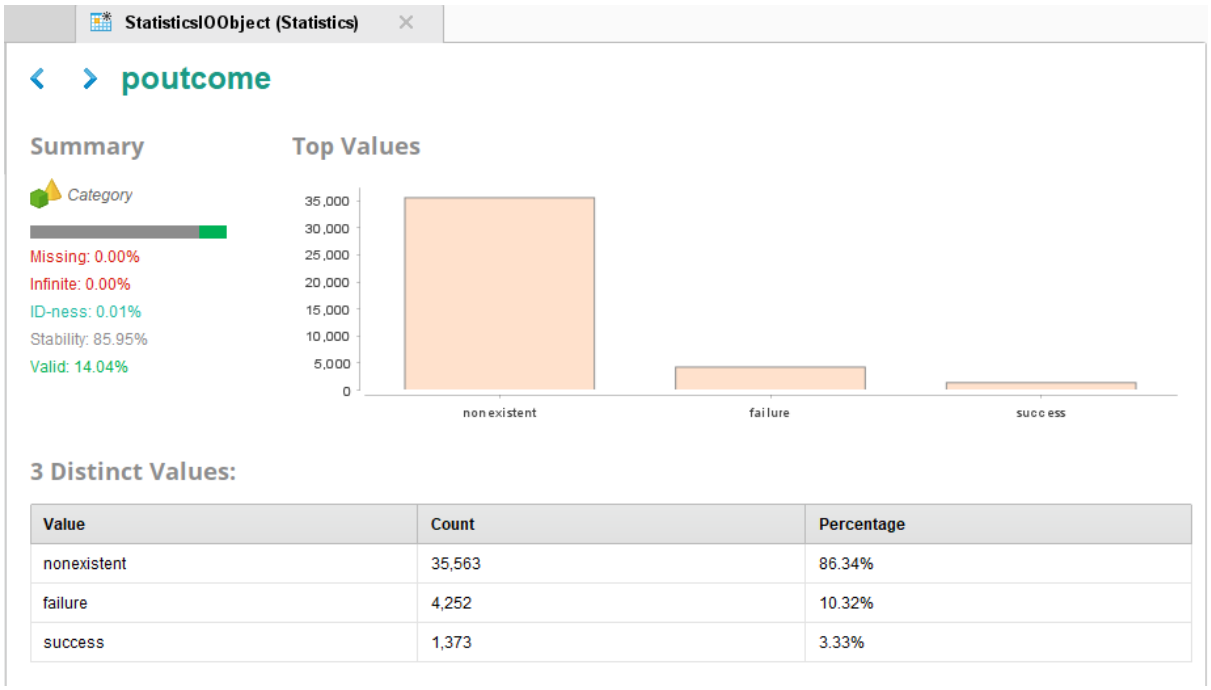
4 Distinct Values:

Value	Count	Percentage
married	24,928	60.52%
single	11,568	28.09%
divorced	4,612	11.20%
unknown	80	0.19%



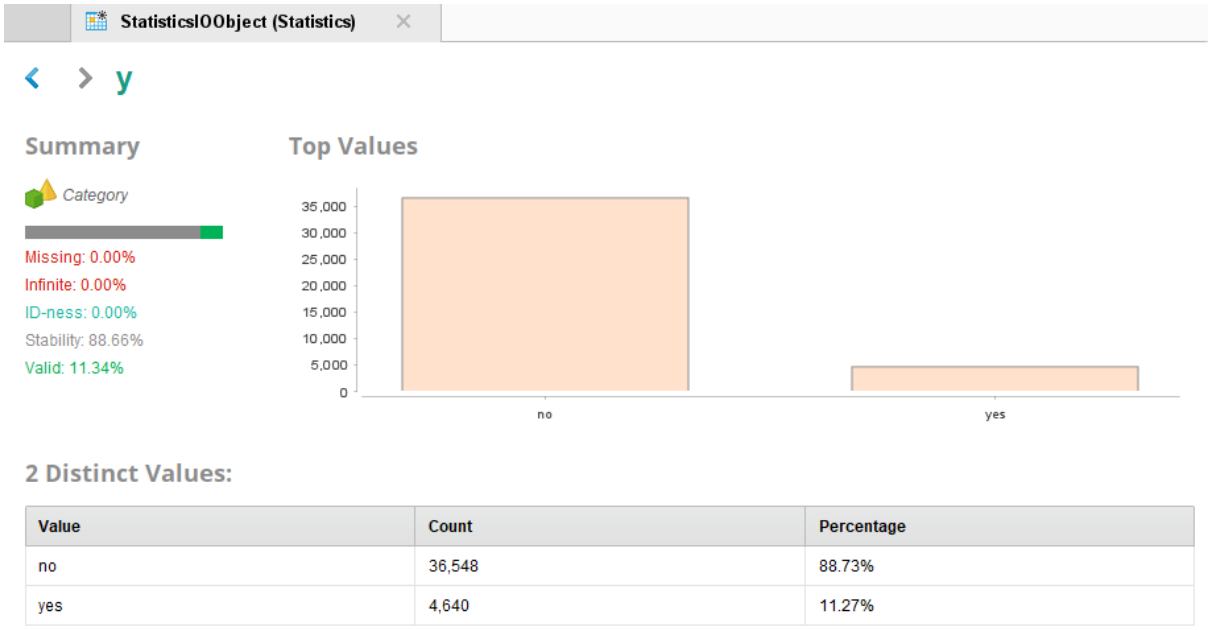
10 Distinct Values:

Value	Count	Percentage
may	13,769	33.43%
jul	7,174	17.42%
aug	6,178	15.00%
jun	5,318	12.91%
nov	4,101	9.96%
apr	2,632	6.39%



3 Distinct Values:

Value	Count	Percentage
nonexistent	35,563	86.34%
failure	4,252	10.32%
success	1,373	3.33%



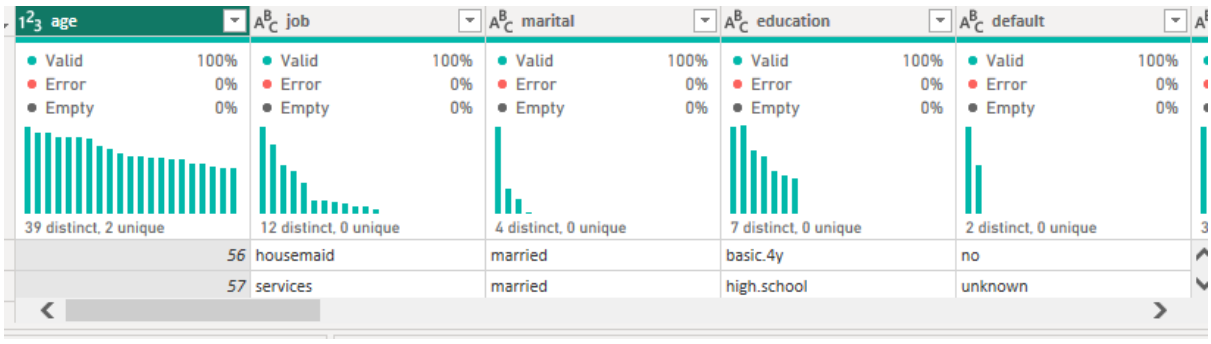
Value	Count	Percentage
no	36,548	88.73%
yes	4,640	11.27%

B)DATA QUALITY ISSUES

Data Quality Issues refer to errors within datasets. Ensuring data quality is a fundamental step in deriving valuable insights from data, whether in traditional data warehousing or newer Big Data environments, and requires dedicated processes, tools, and governance (Ladley, 2020). It is a key indicator of the effectiveness of a Data Governance program. In order to identify and cleanse Data POWER BI was used throughout until a predictive model was made using it.

Some data quality issues include Missing Data, inconsistent Data, Outliers and Anomalies, Data Bias and Skewness, Incorrect Data Types, Duplicate Records, Relevance of Attributes and Data Granularity Issues (Sharda et al,2019, p. 221). The dataset contains no missing values/ has no empties as seen in diagram 1 below.

Diagram 1

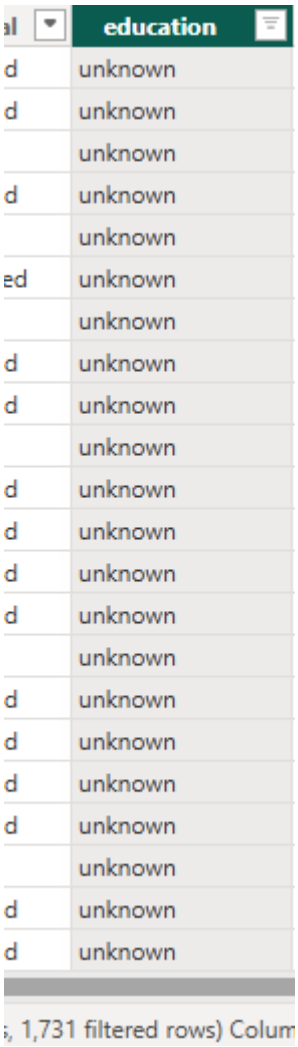


Even though it is so. The dataset still has Outliers & contains unknowns in some

categorical attributes which were possibly placed in attempts to avoid having gaps in the data . Fields with outliers include Job, Marital, Education, Loan, Housing, Default.

Education has 1,731 unknowns as visible in diagram 2 below:

Diagram 2



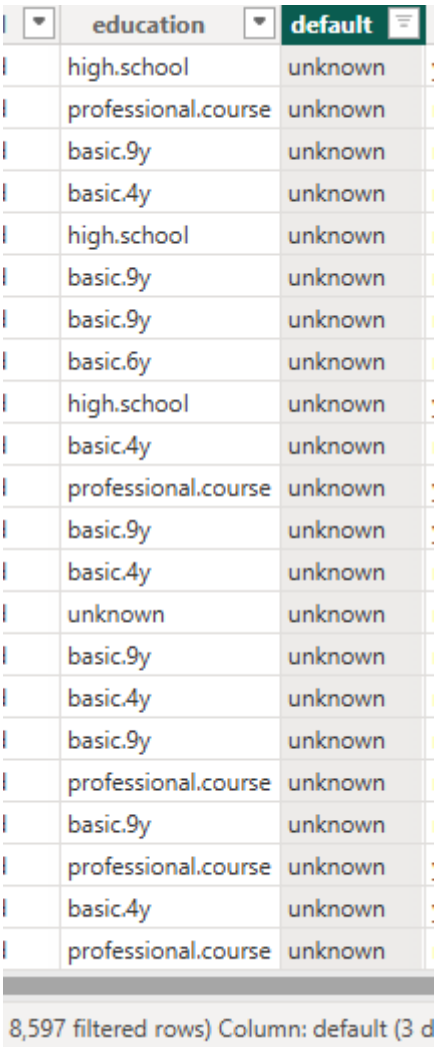
Job has 330 unknowns as visible inn diagram 3 below

job	marital	educatic
unknown	single	university.d
unknown	single	high.schoo
unknown	married	unknown
unknown	divorced	unknown
unknown	married	basic.9y
unknown	married	unknown
unknown	married	university.d
unknown	married	professiona
unknown	single	basic.4y
unknown	married	professiona
unknown	married	unknown
unknown	single	university.d
unknown	single	high.schoo
unknown	single	basic.9y
unknown	married	unknown
unknown	single	university.d
unknown	single	university.d
unknown	married	basic.4y
unknown	married	unknown
unknown	single	unknown
unknown	single	high.schoo
unknown	married	university.d

tional-full (41,188 rows, 330 filtered rows

Diagram 3

Default has 8,597 unknowns as shown in diagram 5

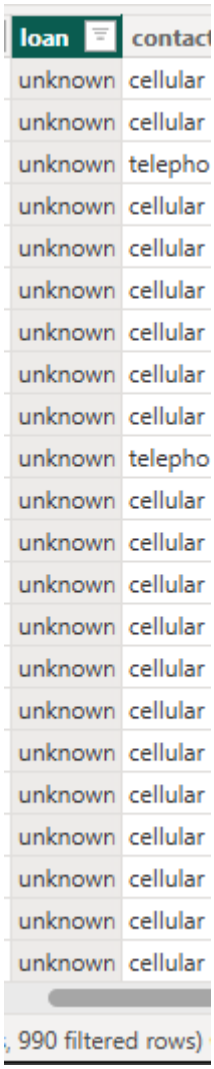


education	default
high.school	unknown
professional.course	unknown
basic.9y	unknown
basic.4y	unknown
high.school	unknown
basic.9y	unknown
basic.9y	unknown
basic.6y	unknown
high.school	unknown
basic.4y	unknown
professional.course	unknown
basic.9y	unknown
basic.4y	unknown
unknown	unknown
basic.9y	unknown
basic.4y	unknown
basic.9y	unknown
professional.course	unknown
basic.9y	unknown
professional.course	unknown
basic.4y	unknown
professional.course	unknown

8,597 filtered rows) Column: default (3 d

Diagram 5

Loan has 990 unknowns as visible in Diagram 6 below



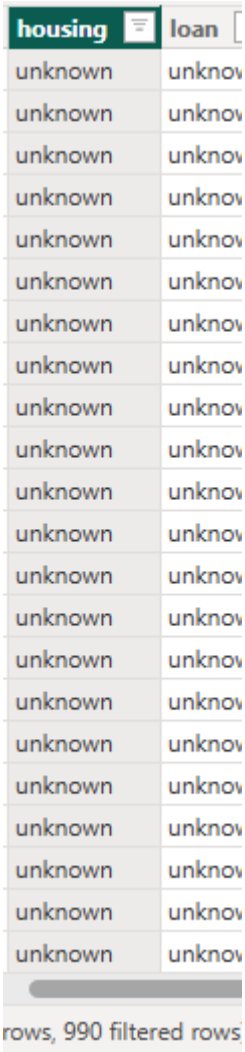
loan	contact
unknown	cellular
unknown	cellular
unknown	telepho
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular
unknown	telepho
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular
unknown	cellular

990 filtered rows)

Diagram 6

Housing has 990 unknowns as seen in Diagram 7 below

Diagram 7



housing	loan
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow
unknown	unknow

rows, 990 filtered rows

Another issue is Relevance of Attributes. The Duration column is irrelevant and can only be included for benchmark purposes and could possibly be discarded if the intention is to have a realistic predictive model which is currently the case. [Moro et al., 2014]

Some names of columns in the dataset are unclear and hard to understand without getting and explanation

c) To deal with the issue of outliers I used the filter text filters option in Power Bi. I defined a filter being does not equal to unknown for the fields Job, Marital, Education, Loan, Housing and Default as visible in Image 1 below. This change left me with 30,488 rows.

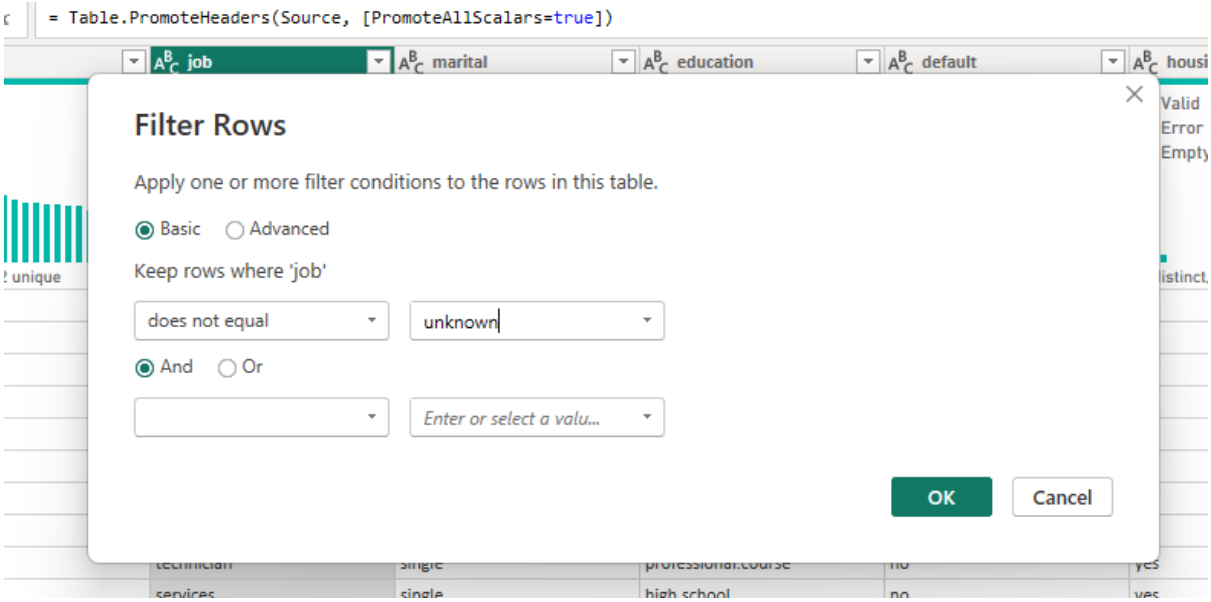


Image 1

In order to ensure Relevance of Attributes & provide accuracy, I discarded the duration column which could possibly ensure that I focus the Predictive model on features known before the call like age which would help make a realistic model that prioritizes clients to call. This change could ensure that the model focuses on actionable and available data, not from outcomes. Image 2 below shows. This change left me with 20 attributes.

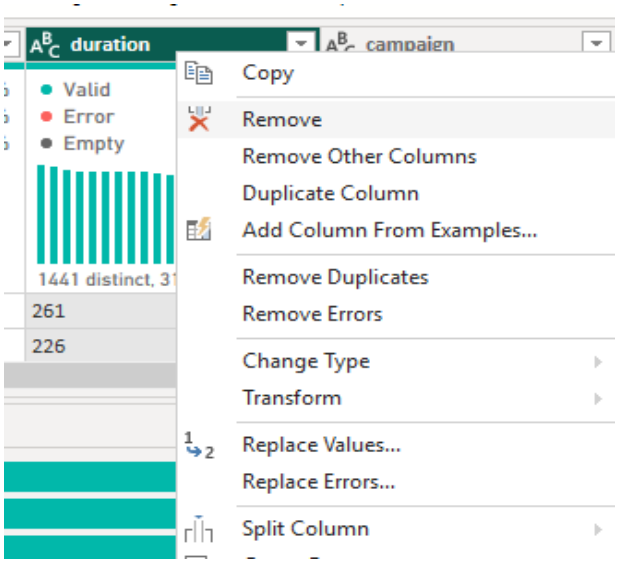

duration
campaign
Valid
Error
Empty
1441 distinct, 31
261
226
Copy
Remove
Remove Other Columns
Duplicate Column
Add Column From Examples...
Remove Duplicates
Remove Errors
Change Type
Transform
Replace Values...
Replace Errors...
Split Column

Image 2

d) DASHBOARD TO PREDICT IF A USER WILL SUBSCRIBE TO TERM Y

The actor for My Dashboard is the Marketing Manager. The visuals I used that shall be informative include:

Slicer for months: Slicers are a common and effective feature in dashboards that allow users to easily filter the entire view based on a selected category, such as month. This interactivity enables the marketing manager to explore how metrics and trends vary depending on the specific month of the campaign, which is crucial context, slicers encourage exploration and deeper analysis.

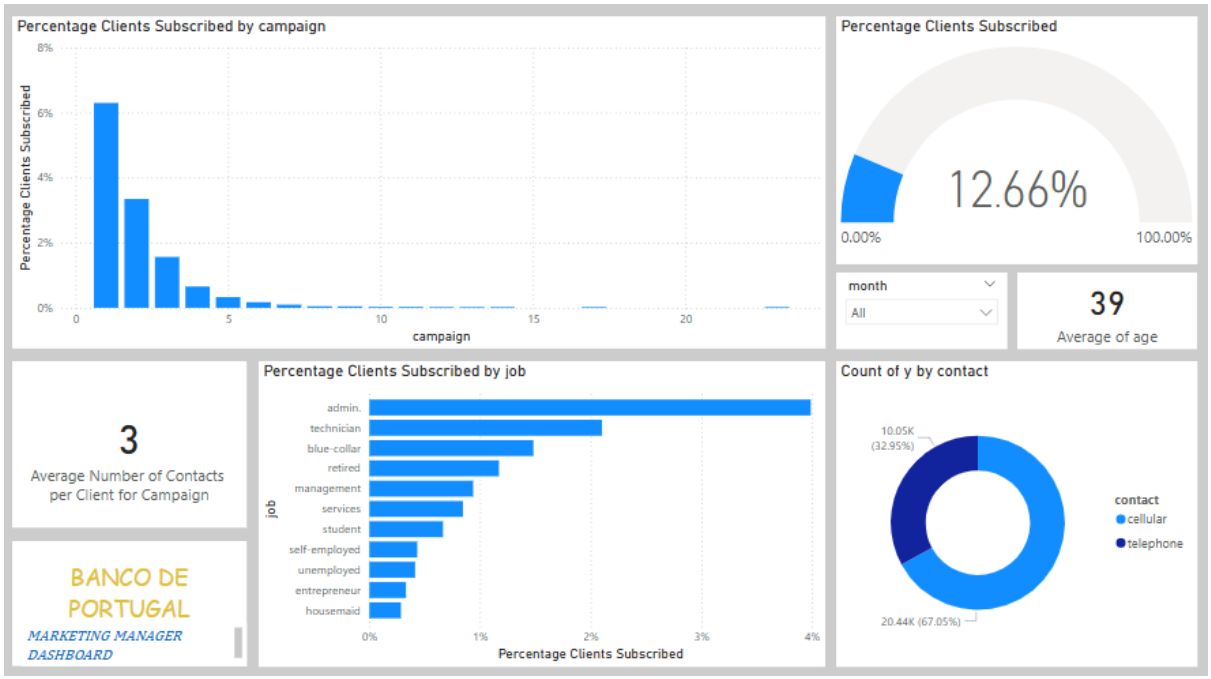
Column chart for Percentage Clients Subscribed by campaign: percentage of clients subscribed plotted against the campaign number (representing the number of contacts). A columnn cgart is explicitly recommended for comparing metrics across discrete categories. Campaign number is a discrete category. Plotting the percentage subscribed for each number of contacts visually highlights how the subscription rate changes with repeated contact, helping to identify potential trends or an optimal number of contacts.

Gauge for Percentage Clients Subscribed: Gauges are specifically for displaying a single, key overall performance metric prominently. They provide a quick, high-level view of the total conversion rate, which is a fundamental measure of success for the campaign and essential for a marketing manager to see at a glance.

Donut chart for Count of Y by contact: Used a donut chart to show the distribution of subscribers ('y') by contact method (cellular vs. telephone). Donut charts (a variation of pie charts) are often used to illustrate the relative proportions of a whole. In this case, it shows the proportion of total subscribers reached by each contact method. Donut charts are a common chart type and can effectively convey a simple part-to-whole relationship, like the breakdown of contacts/subscribers by method.

Bar chart for Percentage Clients Subscribed by Job: Used a horizontal bar chart to display the percentage of clients subscribed across different job types. Similar to the column chart for campaigns, a bar chart is highly recommended for comparing values across discrete categories like job types. Using horizontal bars can be particularly useful when the category labels (job titles) are long. Sorting the bars by the subscription rate, as done in the visual, is a best practice to enhance clarity and make it easy to identify which job types have the highest or lowest rates. This comparison helps reveal which client segments (job types) were most receptive to the campaign.

1.DASHBOARD OVERALL IMAGE:



2. Interpretation of visuals:

In order to attain the reading for Percentage Clients Subscribed I used a DAX formula. Firstly I made a measure for total clients as visible in the image DAX 1 below.

```
Total Clients = COUNTROWS(ALL('bank-additional-full'))
```

DAX 1

I subsequently defined a measure to help me find the total of those who subscribed as visible in the image DAX 2 below.

```
SubscriberNo = CALCULATE(COUNTROWS('bank-additional-full'), FILTER('bank-additional-full', 'bank-additional-full'[y] = "yes"))
```

DAX 2

To make the percentage Clients subscribed. A measure was made from Both DAX 1 & DAX 2 as visible in DAX 3 below.

```
Percentage Clients Subscribed = DIVIDE([SubscriberNo], [Total Clients])
```

DAX 3

The above formula then became the Backbone of the overall Analysis.

Percentage Clients Subscribed by Campaign Shows that Campaign 1 has the highest reading of subscribers (Figure 1 below) and the more times customers are contacted the less likely they are to subscribe. For the Bank this means the marketing team should focus on making a great first impression on the very first contact and focus on new customers every time instead of returning to calling a customer who said no to subscribing before.

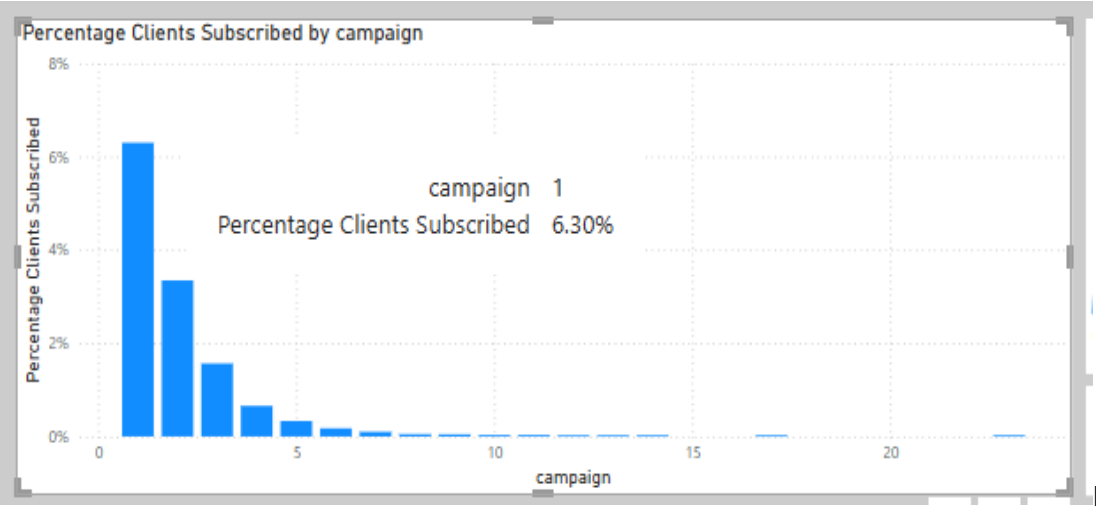


Figure 1

Percentage Clients subscribed by job indicates that the job type admin is the one that subscribes the most(Figure 2 below) making up 3.99% of the total subscribers in the 12.66%. Technicians make up 2.1% ,blue-collar make up 1.48% and those retired make up 1.17%. These readings are the highest and most contributing as they make up over 1%. For the Bank this means the job type types should be the target market and more should be contacted than not, with commercials and advertisements focused on highlighting benefits that subscribing to the term y would provide them.

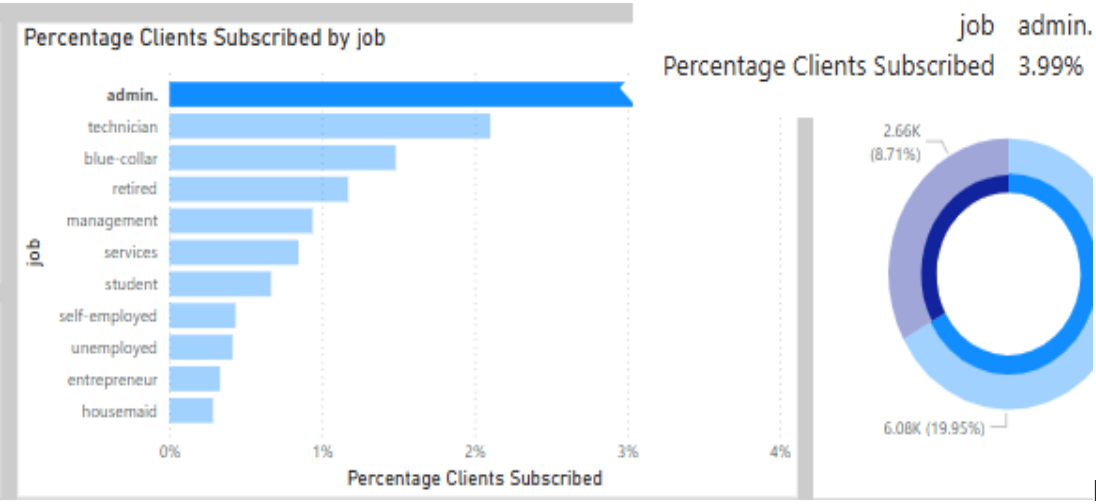


Figure 2

In terms of Count of y by contact method it is visible that more people are contacted using cellular and those contacted by telephone are almost half those contacted by cellular. Those contacted by cellular are more responsive as they make up 10.74% of the grand total of those who subscribed which is 12.66% (Figure 3 below). For the bank this means more energy and effort should be exerted in contacting those using cellular as even more admins(which is the most subscribing job type) use cellular according to the visual.

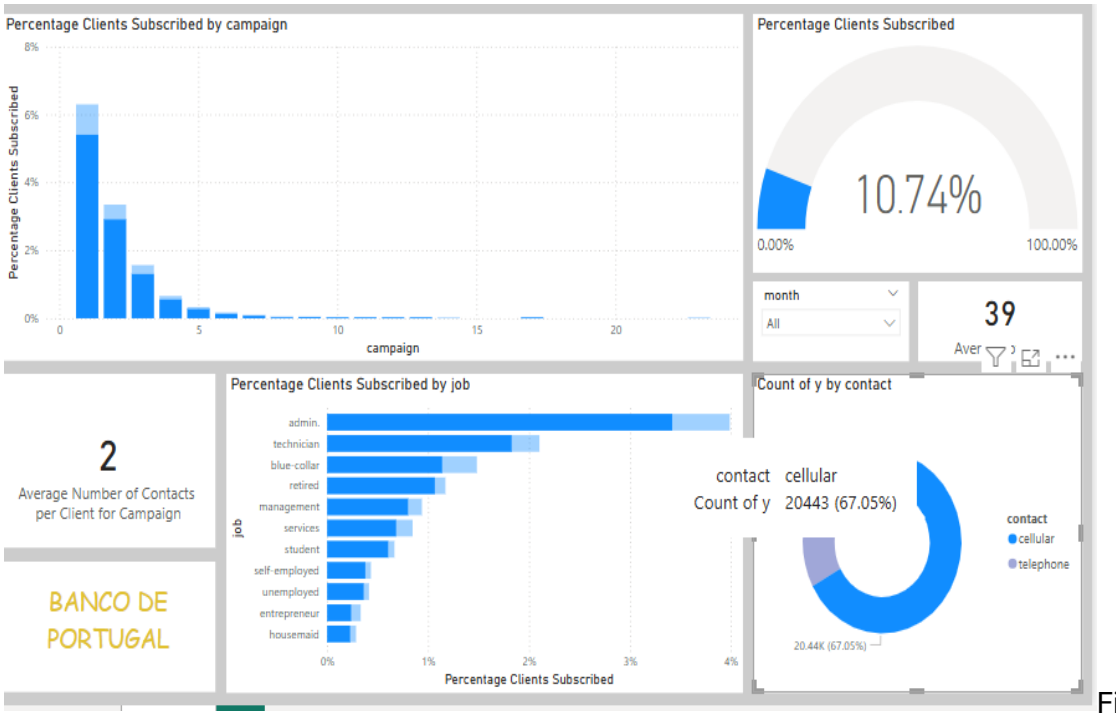


Figure 3

When I use the slicer to filter by months it is observed that the month August has brought the highest number of subscribers which is 1.75% of the grand total of subscribed which is 12.66% (Figure 4 below) . For Marketing this indicated that more energy should be exerted during this month, with more campaigns carried out to take advantage of peoples eagerness to subscribe and spend during August which features a number of holidays including International day of the Worlds Indigenous Peoples and National Woman's Day.

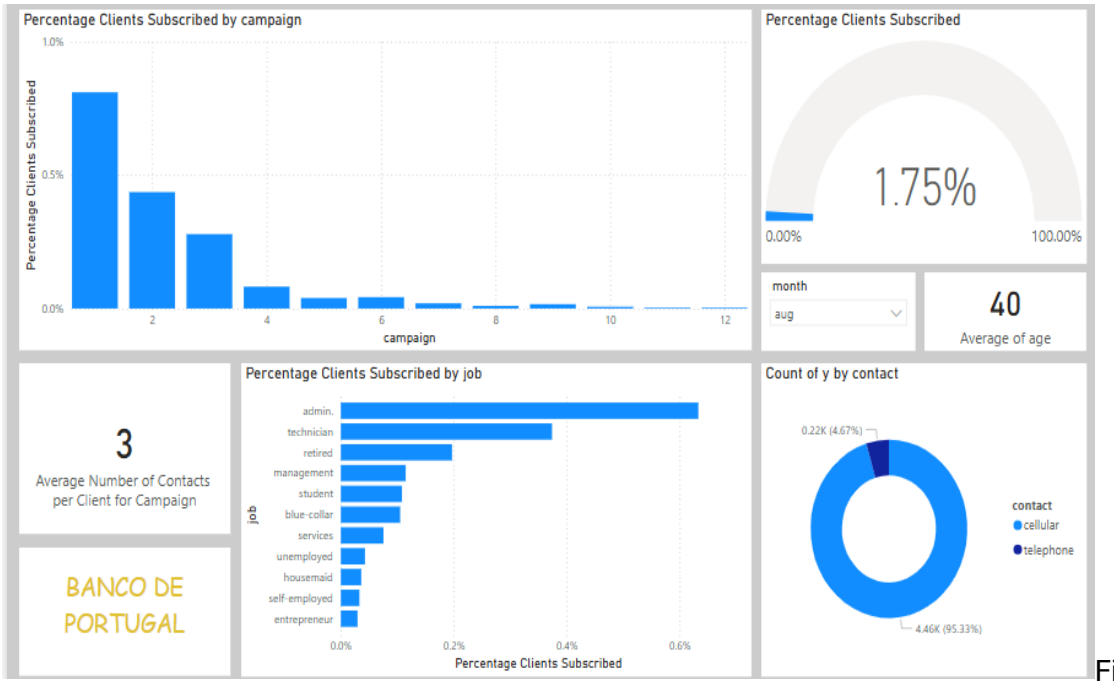


Figure 4

IN CONCLUSION, the data shows that the Bank is not doing well as the dataset contains 30,488 people and yet only 12% has subscribed .This is an unacceptable number as it is of campaigns held for a period of 2 years, which should have produced more palatable results considering the bank has been repeatedly calling the same pool of un-converted clients for 2 full years. I advice the bank to focus on campaigns that highlight importance of the term to the working class which includes at the top, admins. As I reflect I see there is opportunity to improve the initial approaches to customers as there are customers who subscribe after being contacted more than once which means they would get convinced after hearing a few other details.

REFERENCE LIST

Ladley, J., 2019. *Data governance: how to design, deploy, and sustain an effective data governance program*. 2nd ed. Amsterdam: Morgan Kaufmann. Available at: https://www.econbiz.de/Record/data-governance-how-to-design-deploy-and-sustain-an-effective-data-governance-program-ladley-john/10012150794 [Accessed 10 May 2025]

Sharda, R., Delen, D. and Turban, E. (2019) Business Intelligence: A Managerial Perspective on Analytics, 3rd edn. Harlow, Pearson Education

S. Moro, P. Cortez and P. Rita. A Data-Driven Approach to Predict the Success of Bank Telemarketing. Decision Support Systems, In press, http://dx.doi.org/10.1016/j.dss.2014.03.001

UCI Machine Learning Repository, 2025. *UCI Machine Learning Repository*. [online] University of California, Irvine. Available at: https://archive.ics.uci.edu/ [Accessed 8 May 2025]